



RESIT EXAM OF: ALGEBRA 1

Duration: 1h 30min

➡ **Answers must be organised, clear, and justified.**

--- **Exercise 1** (4 points) -----

1. Indicate, with justification, if the following statement is true or false, then give its negation.

$$\forall n \in \mathbb{N}, \exists p \in \mathbb{N}, n(n+1) = 2p.$$

2. Using the proof by contradiction, prove that $\forall m \in \mathbb{Z}$:

$$m \text{ is odd} \Rightarrow m^2 - 5 \text{ is even.}$$

--- **Exercise 2** (7 points) -----

I. Let $\mathcal{R}$ be a binary relation defined on $\mathbb{Z}$ as follows:

$$a \mathcal{R} b \Leftrightarrow \exists k \in \mathbb{Z}, a - b = 5k.$$

1. Is $\mathcal{R}$ antisymmetric? Justify.

2. Is $\mathcal{R}$ an equivalence relation? Justify.

II. Let f be a function defined from $\mathbb{R}$ to $\mathbb{R}$ by: $f(x) = \frac{4x}{x^2+1}$.

1. Show that: $\forall \alpha \in \mathbb{R}^*, f(\alpha) = f\left(\frac{1}{\alpha}\right)$.

2. Determine and $f^{-1}(\{2, 3\})$.

3. Is f injective? Surjective? Bijective? Justify.

--- **Exercise 3** (5 points) -----

Let $z = 1 + i$ and $w = -1 + i\sqrt{3}$ be two complex numbers.

1. Determine the modulus and an argument to each z and w .

2. Express z and w in exponential forms.

3. Determine the algebraic and polar forms of $\frac{z}{w}$.

4. Deduce $\cos \frac{-5\pi}{12}$ and $\sin \frac{-5\pi}{12}$.

5. Solve the following equation in $\mathbb{C}$: $(z - 1)^3 + i = 0$.

--- **Exercise 4** (4 points) -----

Let $*$ be an operation defined on a set $G = \mathbb{R} - \{-3\}$ as follows: $x * y = xy + 3(x + y) + 6$.

Show that $(G, *)$ is an abelian group.

----- Good luck -----